

Arduino for Embedded Systems - Beginner Level

Lesson 1: Introduction to Arduino and Development Environment

Introduction to Arduino and Development Environment – Study Notes

Key Concepts

1. **Arduino board anatomy** – power jack, USB port, microcontroller, digital & analog I/O pins, power pins, reset button, LED indicators.
2. **Arduino IDE installation** – supported operating systems, driver installation (Windows), board manager, selecting the correct board and port.
3. **Sketch structure** – `setup()` and `loop()` functions, basic syntax, uploading code to the board.
4. **The “Blink” sketch** – the classic first program, using `digitalWrite()` and `delay()` to toggle the on board LED.
5. **Debugging basics** – serial monitor, common upload errors, checking wiring and board selection.

Summary

The Arduino platform provides a simple, low cost way to prototype electronic projects. An Arduino board is essentially a small computer built around a microcontroller (e.g., ATmega328P on the Uno). Understanding the board's physical layout—where power enters, how the USB connection supplies both programming data and power, and which pins are used for digital and analog I/O—is the first step toward building circuits.

To write and upload programs (called *sketches*) you use the Arduino Integrated Development Environment (IDE). After downloading the IDE for your OS, you may need to install a USB driver (Windows) and then select the appropriate board type and COM port. A sketch always contains two mandatory functions: `setup()`, which runs once at start up, and `loop()`, which repeats indefinitely. The classic “Blink” example demonstrates the workflow: open the example, verify the code, upload it, and watch the on board LED (pin 13) toggle on and off. This simple experiment confirms that the hardware, software, and communication chain are all working correctly, giving you a solid foundation for more complex projects.

Important Points

- **Board parts**
 - **Power jack** – external supply (7 12 V).
 - **USB port** – program upload & serial communication; also powers the board.
 - **Microcontroller** – executes the sketch (e.g., ATmega328P).
 - **Digital I/O pins (0 13)** – can read/write HIGH/LOW.
 - **Analog input pins (A0 A5)** – 10 bit ADC for sensor values.
 - **Power pins** – 5 V, 3.3 V, GND, VIN.

- **Reset button** – restarts the microcontroller.
- **LEDs** – **ON** (power), **L** (pin /13 built in LED), **TX/RX** (serial activity).
 - **IDE installation steps**
 1. Download the latest Arduino IDE from [/\[arduino.cc\]\(https://www.arduino.cc/en/software\)](https://www.arduino.cc/en/software).
 2. Run the installer (Windows/macOS) or extract the package (Linux).
 3. On Windows, install the **Arduino USB driver** when prompted.
 4. Launch the IDE !' **Tools > Board > Arduino Uno** (or your board).
 5. **Tools > Port** !' select the COM port that appears when the board is plugged in.
 - **Sketch anatomy**

```

```cpp
void setup() { // runs once
 pinMode(13, OUTPUT); // set LED pin as output
}
void loop() { // runs repeatedly
 digitalWrite(13, HIGH);
 delay(1000); // wait 1 second
 digitalWrite(13, LOW);
 delay(1000);
}
```

```

- **Uploading** – Click the **Verify** (check mark) button !' compile; then click **Upload** (right arrow) !' IDE sends the compiled binary via the bootloader.
- **Common issues**
 - “avrdude: stk500_recv(): programmer is not responding” !' wrong board/port or missing driver.
 - No LED blink !' board not powered, wrong pin number, or sketch not uploaded.

Examples

1. Blink – the “Hello World” of Arduino

1. Open **File > Examples > 01.Basics > Blink**.
2. Verify the code (✓).
3. Connect an Arduino Uno via USB.
4. Select the correct board and port.
5. Upload (!).

What happens? The on board LED on pin /13 flashes once per second, proving that the IDE can compile, transfer, and run code on the board.

2. External LED Blink

Hardware

- LED + 220 Ω & W6—7F ÷ .
- Connect the resistor to **digital pin /8**, LED anode to the resistor, LED cathode to GND.

Code

```

```cpp

```

```

const int ledPin = 8; // pin where LED is connected
void setup() {
 pinMode(ledPin, OUTPUT);
}
void loop() {
 digitalWrite(ledPin, HIGH); // LED on
 delay(500); // half second
 digitalWrite(ledPin, LOW); // LED off
 delay(500);
}
...

```

**\*\*Explanation\*\*** – This shows how to control any digital pin, not just the built in LED, and reinforces wiring a simple external circuit.

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## Quick Review

1. **\*\*Name three major sections of an Arduino board and their purpose.\*\***
2. **\*\*What two functions must every Arduino sketch contain, and when does each run?\*\***
3. **\*\*Describe the steps required to install the Arduino IDE on a Windows PC.\*\***
4. **\*\*In the Blink sketch, which command sets a pin as an output and which command changes its state? \*\***
5. **\*\*If the IDE reports “programmer is not responding,” list two possible causes and how to fix them.\*\***

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## Further Reading

- **\*\*Arduino board families\*\*** – Uno, Nano, Mega, Leonardo, MKR series; differences in microcontroller and I/O count.
- **\*\*Serial communication\*\*** – Using ``Serial.begin()``, the Serial Monitor, and sending data to a PC.
- **\*\*Analog input & PWM output\*\*** – ``analogRead()``, ``analogWrite()``, and using sensors/actuators.
- **\*\*Using libraries\*\*** – Installing and including external libraries (e.g., ``Servo``, ``Wire``).
- **\*\*Basic electronics for Arduino\*\*** – Resistors, LEDs, buttons, pull up/down resistors, breadboarding techniques.

Happy prototyping! Ø=Þ€

